

Yichao Cai, Ph.D. Candidate in Computer Science

CONTACT INFORMATION	Australian Institute for Machine Learning Adelaide University Adelaide, SA 5005 Australia	Telephone: +61 478 927 202 E-mail: yichao.cai@adelaide.edu.au yichaocai.com GitHub Google Scholar
RESEARCH INTERESTS	Representation Learning; Multimodal and Multi-View Learning; Identifiability Analysis; Responsible and Explainable AI.	
RESEARCH STATEMENT	My research studies how modern learning objectives and supervision signals shape learned representations. I am particularly interested in when these objectives identify causal and generalizable latent structure, and when they instead discard, conflate, or leave it under-determined. This perspective helps characterize the limits of foundation model objectives and distinguish capabilities that emerge through scaling from those requiring new objectives, supervision, or data interventions. Methodologically, I use tools from identifiability theory, latent-variable modeling, population-objective analysis, and representation geometry. My broader goal is to develop a theory of representation learning that explains the capabilities and structural limits of modern learning systems.	
EDUCATION	Doctor of Philosophy Computer Science, Adelaide University Advisor: Prof. Javen Qinfeng Shi Research: representation learning, identifiability theory, and representation geometry Master in Science Instrument Science and Technology, Wuhan University of Technology Advisor: Prof. Xiao Zhou Research: image processing, computer vision, and autonomous-driving perception Bachelor in Engineering Measurement and Control Technology and Instrumentation, Wuhan University of Technology	Sep 2023–Jun 2027 Sep 2016–Jun 2019 Sep 2012–Jun 2016
PUBLICATIONS	FIRST-AUTHOR / CO-FIRST-AUTHOR Cai, Y. and Shi, J. Q. “On the Identifiability of Masked Prediction: Mode Blindness and Mask Schedules.” <i>arXiv:2608.01383</i>, 2026 (Preprint). Cai, Y.; Zhang, Z.; Liu, Y.; and Shi, J. Q. “The Geometric Mechanics of Contrastive Representation Learning: Alignment Potentials, Entropic Dispersion, and Cross-Modal Divergence.” <i>International Conference on Machine Learning (ICML)</i>, 2026. Cai, Y.*; Liu, Y.*; Gao, E.; Jiang, T.; Zhang, Z.; van den Hengel, A.; and Shi, J. Q. “On the Value of Cross-Modal Misalignment in Multimodal Representation Learning.” <i>Advances in Neural Information Processing Systems (NeurIPS)</i>, 2025. Spotlight. *Equal contribution. Cai, Y.; Liu, Y.; Zhang, Z.; and Shi, J. Q. “CLAP: Isolating Content from Style Through Contrastive Learning with Augmented Prompts.” <i>European Conference on Computer Vision (ECCV)</i>, 2024. COLLABORATIVE - Liu, Y.; Gong, D.; Cai, Y.; Gao, E.; Zhang, Z.; Huang, B.; Gong, M.; van den Hengel, A.; and Shi, J. Q. “I Predict Therefore I Am: Is Next Token Prediction Enough to Learn Human-Interpretable Concepts from Data?” <i>International Conference on Learning Representations (ICLR)</i>, 2026. - Jiang, W.; Liu, Y.; Cai, Y.; Gao, E.; Dong, J.; Abbasnejad, E.; Yao, L.; and Shi, J. Q. “What Makes a Representation Good for Single-Cell Perturbation Prediction?” <i>International Conference on Machine Learning (ICML)</i>, 2026. - Chen, J.; Yin, Z.; Cai, Y.; Liu, Y.; Zhang, Z.; Gong, D.; and Shi, J. Q. “Boundary Embedding Shaping with Adaptive Contrastive Learning for Graph Structural Disentanglement.” <i>International Conference on Machine Learning (ICML)</i>, 2026.	

INVITED TALKS	Understanding multimodal contrastive learning through two lenses: identifiability and population geometry Vision & Sensing Lab, Technion; Online invited talk EML, Helmholtz Munich; Online invited talk	Jul 2026
TEACHING	At Adelaide University (formerly the University of Adelaide): Teaching Assistant, Programming for Artificial Intelligence Teaching Assistant, Neural Networks and Deep Learning Guest Lecturer & Head Tutor, Statistical Machine Learning Teaching Assistant, Using Machine Learning Tools Teaching Assistant, Concepts in AI and ML	Semester 2, 2026 Semester 1, 2026 Semester 2, 2025 Trimester 2, 2025 Semester 1, 2025
ACADEMIC SERVICE	Conference Reviewing: AAAI 2027; NeurIPS 2026; ICML 2026 (Silver Reviewer Award); ICLR 2026. Journal Reviewing: Transactions on Machine Learning Research (TMLR).	
HONORS & AWARDS	The D. R. Stranks Travelling Fellowship NeurIPS 2025 Scholar Award, Neural Information Processing Foundation University of Adelaide Research Scholarship Outstanding Graduate Student, Wuhan University of Technology	Jun 2026 Oct 2025 Sep 2023 Jun 2019
VISITING EXPERIENCE	Visiting Student Researcher California PATH, UC Berkeley – Conducted research on visual perception algorithms for autonomous driving.	May 2018–Sep 2018